

Covariance__Correlation

Math 742

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Joint Distributions

Definition 1: For a pair of discrete random variables X and Y , the joint mass function $f(x, y) = \mathbb{P}(X = x \text{ and } Y = y)$.

For 2 variables, you can generally write the joint PMF with a table.

Example: Let X and Y be two discrete random variables with joint PMF as follows:

$f(x, y)$	$Y = 0$	$Y = 1$	$Y = 2$	Total
$X = 0$	1/9	2/9	3/9	6/9
$X = 1$	0	1/9	2/9	3/9
Total	1/9	3/9	5/9	1

So, $f(1, 1) = 1/9$

Definition 2: For a pair of continuous random variables X and Y , we call the function $f(x, y)$ a joint PDF for the random variables (X, Y) if:

- $f(x, y) \geq 0$ for all (x, y)
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$
- For any set $A \subset \mathbb{R} \times \mathbb{R}$, $\mathbb{P}((X, Y) \in A) = \int \int_A f(x, y) dx dy$

In either case, the **joint CDF** is $F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y)$

Definition 3: For discrete random variables with joint PMF $f(x, y)$, the **marginal** PMF for X is determined by,

- $f_X(x) = \mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x, Y = y) = \sum_y f(x, y)$
- $f_Y(y) = \mathbb{P}(Y = y) = \sum_x \mathbb{P}(X = x, Y = y) = \sum_x f(x, y)$

And if X and Y are continuous,

- $f_X(x) = \int f(x, y) dy$ and $f_Y(y) = \int f(x, y) dx$

Example: Consider the previous example:

$f(x, y)$	$Y = 0$	$Y = 1$	$Y = 2$	Total
$X = 0$	1/9	2/9	3/9	6/9
$X = 1$	0	1/9	2/9	3/9
Total	1/9	3/9	5/9	1

X	$f_X(x)$
0	6/9
1	3/9

Y	$f_Y(y)$
0	1/9
1	3/9
2	5/9

Example: Let X and Y be two continuous random variables with the following joint PDF:

$$f(x, y) = \begin{cases} c(x^2 + y) & 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Determine the value of c that will make this a PDF.

$$\int_0^2 \int_0^1 c(x^2 + y) dy dx = 1$$

Inside integral:

$$\begin{aligned} & \int_0^1 c(x^2 + y) dy \\ &= \int_0^1 cx^2 dy + \int_0^1 cy dy = cx^2 y \Big|_0^1 + \frac{cy^2}{2} \Big|_0^1 \\ &= cx^2 + \frac{1}{2}c \end{aligned}$$

Now, integrate this with respect to x (i.e. outside integral),

$$\begin{aligned} & \int_0^2 \left(cx^2 + \frac{1}{2}c \right) dx \\ &= \frac{cx^3}{3} \Big|_0^2 + \frac{1}{2}cx \Big|_0^2 \\ &= \frac{8c}{3} + c \end{aligned}$$

Now set this equal to 1 and solve for c ,

$$\frac{8c}{3} + c = 1$$

$$\frac{11c}{3} = 1$$

$$c = \frac{3}{11}$$

2. Determine the marginal PDF for X and the marginal PDF for Y .

$$\begin{aligned} f_X(x) &= \int_0^1 \frac{3}{11}(x^2 + y)dy \\ &= \frac{3}{11} \left(x^2 y + \frac{y^2}{2} \right) \Big|_0^1 = \frac{3}{11} \left(x^2(1) + \frac{1^2}{2} \right) = \frac{3}{11}(x^2 + 1/2) \\ f_X(x) &= \begin{cases} \frac{3}{11}(x^2 + 1/2) & : 0 \leq x \leq 2 \\ 0 & : \text{otherwise} \end{cases} \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \int_0^2 \frac{3}{11}(x^2 + y)dx \\ &= \frac{3}{11} \left(\frac{x^3}{3} + yx \right) \Big|_0^2 \\ &= \frac{3}{11} \left(\frac{8}{3} + 2y \right) - 0 \\ &= \frac{8}{11} + \frac{6}{11}y \end{aligned}$$

$$f_Y(y) = \begin{cases} \frac{8}{11} + \frac{6}{11}y & : 0 \leq y \leq 1 \\ 0 & : \text{otherwise} \end{cases}$$

Definition 4: The **conditional** PMF is:

$$f_{X|Y}(x|y) = \mathbb{P}(X = x|Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

Provided, $f_Y(y) > 0$.

Definition 5: For continuous random variables X and Y the **conditional** probability density function is:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

This assumes that $f_Y(y) > 0$. Then,

$$\mathbb{P}(X \in A|Y = y) = \int_A f_{X|Y}(x|y)dx$$

Example: For the previous joint PDF, determine $\mathbb{P}(X < 1/2|Y = 1/4)$.

First find the conditional distribution:

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{\frac{3}{11}(x^2 + y)}{\frac{8}{11} + \frac{6}{11}y} \\ \mathbb{P}(X < 1/2|Y = 1/4) &= \int_0^{1/2} \frac{\frac{3}{11}(x^2 + 1/4)}{\frac{8}{11} + \frac{6}{11}(1/4)} dx \\ &= \int_0^{1/2} \frac{\frac{3}{11}(x^2 + 1/4)}{19/22} \end{aligned}$$

$$\begin{aligned}
&= \frac{22}{19} \frac{3}{11} \int_0^{1/2} (x^2 + 1/4) \\
&= \frac{6}{19} \left(\frac{x^3}{3} + \frac{1}{4}x \right) \Big|_0^{1/2} = \frac{6}{19} \left(\frac{1}{24} + \frac{1}{8} \right) = \frac{6}{19} \left(\frac{1}{6} \right) = \frac{1}{19}
\end{aligned}$$

In the previous example, are X and Y independent? If so, $f(x, y) = f_X(x)f_Y(y)$.

$$f_X(x)f_Y(y) = \frac{3}{11}(x^2 + 1/2) \left(\frac{8}{11} + \frac{6}{11}y \right)$$

Clearly, this does not equal: $\frac{3}{11}(x^2 + y)$, so X and Y are not independent.

Definition 6: If $X_1, X_2, \dots, X_n$ are independent random variables and each has the same marginal distribution with CDF F , we say that $X_1, X_2, \dots, X_n$ are IID (*independent and identically distributed*) and we write,

$$X_1, X_2, \dots, X_n \sim F$$

If F has probability density function f , we can also write, $X_1, X_2, \dots, X_n \sim f$ and we can say that $X_1, X_2, \dots, X_n$ is a random sample of size n from F .

Definition 7: Let X and Y be random variables with mean of X equal to μ_X and mean of Y equal to μ_Y . Also, the standard deviation of X is σ_X and standard deviation of Y σ_Y . The **covariance** between X and Y is defined as,

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

and the correlation as,

$$\rho_{X,Y} = \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

You can think of covariance and correlation is a measure of how associated two random variables are. If on average, when X is higher than its mean Y is also higher than its mean and likewise, when X is lower than its mean Y tends to be lower than its mean this would give us a positive covariance. The reverse would give us a negative covariance.

Theorem: $\text{Cov}(X, Y) = \mathbb{E}(XY) - \mu_X \mu_Y$.

Proof: Let's assume X and Y are discrete. For the continuous case, replace summation signs with integrals.

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

$$= \mathbb{E}[XY - \mu_X X - \mu_Y Y + \mu_X \mu_Y]$$

because μ_X and μ_Y are constant,

$$= \mathbb{E}(XY) - \mu_X \mathbb{E}(X) - \mu_Y \mathbb{E}(Y) + \mu_X \mu_Y$$

$$= \mathbb{E}(XY) - \mu_X \mu_X - \mu_Y \mu_Y + \mu_X \mu_Y$$

$$= \mathbb{E}(XY) - \mu_X \mu_Y$$

There are two problems with covariance and that is why we typically use correlation instead of covariance.

1. Covariance is not scale free, if you change the scale you change the covariance, but you haven't changed the degree of association.
2. Covariance is unbounded. On the other hand, correlation is always between -1 and 1.

Theorem: The correlation coefficient (i.e. ρ) is bounded above by 1 and below by -1. Furthermore, if $Y = aX + b$, $\rho_{X,Y} = 1$ if $a > 0$ and $\rho_{X,Y} = -1$ if $a < 0$.

Proof: First, we must show that $\mathbb{E}[(XY)^2] \leq \mathbb{E}(X^2)\mathbb{E}(Y^2)$. Consider the random variable $X - tY$, t is a real valued constant and X and Y have finite variance (i.e. $\mathbb{V}(X) < \infty$ and $\mathbb{V}(Y) < \infty$).

$$0 \leq \mathbb{E}[(X - tY)^2] = \mathbb{E}(X^2) - 2\mathbb{E}(XY)t + t^2\mathbb{E}(Y^2)$$

Notice that this is a quadratic in t ($at^2 + bt + c$), where $a = \mathbb{E}(Y^2)$, $b = -2\mathbb{E}(XY)$ and $c = \mathbb{E}(X^2)$.

In order for this quadratic to be non-negative it must have non-positive discriminant. In other words,

$$\begin{aligned} b^2 - 4ac &= (-2\mathbb{E}(XY))^2 - 4\mathbb{E}(Y^2)\mathbb{E}(X^2) \leq 0 \\ &= 4\mathbb{E}(XY)^2 - 4\mathbb{E}(Y^2)\mathbb{E}(X^2) \leq 0 \\ \mathbb{E}(XY)^2 &\leq \mathbb{E}(X^2)\mathbb{E}(Y^2) \end{aligned}$$

We can thus take the square root of both sides and we wind up with the Cauchy-Schwarz inequality for random variables. (More on the Cauchy-Schwarz inequality later)

$$|\mathbb{E}(XY)| \leq \sqrt{\mathbb{E}(X^2)\mathbb{E}(Y^2)}$$

Now,

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

Let, $U = X - \mu_X$ and $V = Y - \mu_Y$. Using the Cauchy-Schwarz inequality, we have

$$|\text{Cov}(X, Y)| = |\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]| = |\mathbb{E}(UV)| \leq \sqrt{\mathbb{E}[U^2]\mathbb{E}[V^2]}$$

Notice that $\mathbb{E}[U^2] = \mathbb{V}(X) = \sigma_X^2$, and $\mathbb{E}[V^2] = \mathbb{V}(Y) = \sigma_Y^2$

So in the end, we get $|\text{Cov}(X, Y)| \leq \sigma_X \sigma_Y$.

Finally,

$$|\rho_{X,Y}| = \left| \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \right| \leq 1$$

$$\text{or, } -1 \leq \rho_{X,Y} \leq 1$$

.

To show equality on the end points, let $Y = ax + b$.

$$\text{Cov}(X, Y) = \text{Cov}(X, aX + b) = \mathbb{E}[X(aX + b)] - \mu_X \mu_{aX+b}$$

$$\begin{aligned}
&= \mathbb{E}[aX^2 + bX] - \mu_X(a\mu_X + b) \\
&= 2\mathbb{E}(aX^2) + b\mathbb{E}(X) - a\mu_X^2 - b\mu_X \\
&\quad a(\mathbb{E}[X^2] - \mu_X^2) + b\mu_X - b\mu_X \\
&= a\mathbb{V}(X)
\end{aligned}$$

This makes the correlation,

$$\begin{aligned}
\rho &= \frac{a\mathbb{V}(X)}{\sqrt{\mathbb{V}(X)\mathbb{V}(Y)}} = \frac{a\mathbb{V}(X)}{\sqrt{\mathbb{V}(X)\mathbb{V}(aX + b)}} \\
&= \frac{a\mathbb{V}(X)}{\sqrt{\mathbb{V}(X)a^2\mathbb{V}(X)}} = \frac{a\mathbb{V}(X)}{|a|\mathbb{V}(X)} = \frac{a}{|a|}
\end{aligned}$$

So, $a/|a| = 1$ if $a > 0$ otherwise if $a < 0$, $a/|a| = -1$

Example: Given the following bivariate PDF, determine ρ .

$$f(x, y) = \begin{cases} \frac{21}{4}(x^2y) & x^2 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

First note, that if $x^2 \leq y \leq 1$, then $-1 \leq x \leq 1$.

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mu_X\mu_Y$$

First we will find $f_X(x)$ and $f_Y(y)$.

$$\begin{aligned}
f_X(x) &= \int_{y=x^2}^{y=1} \frac{21}{4}(x^2y)dy \\
&= \frac{21}{4}x^2 \frac{y^2}{2} \Big|_{x^2}^1 = \frac{21}{8}x^2 \left(\frac{1}{2} - x^4 \right)
\end{aligned}$$

Notice that for $0 \leq y \leq 1$, the x -range is, $-\sqrt{y} \leq x \leq \sqrt{y}$.

$$\begin{aligned}
f_Y(y) &= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{21}{4}(x^2y)dx \\
&= \frac{21}{4}y \frac{x^3}{3} \Big|_{-\sqrt{y}}^{\sqrt{y}} = \frac{21}{4} \frac{y}{3} (y^{3/2} - (-y^{3/2})) \\
&= \frac{21}{4} \frac{y}{3} (2y^{3/2}) = \frac{7}{2}y^{5/2}
\end{aligned}$$

$$\mathbb{E}(X) = \int_{-1}^1 x \frac{21}{8}x^2 \left(\frac{1}{2} - x^4 \right) dx = 0$$

$$\mathbb{E}(X^2) = \int_{-1}^1 x^2 \frac{21}{8}x^2 \left(\frac{1}{2} - x^4 \right) dx = \frac{7}{15}$$

$$\mathbb{V}(X) = \frac{7}{15}$$

$$\mathbb{E}(Y) = \int_0^1 y \frac{7}{2} y^{5/2} dy = \frac{7}{9}$$

$$\mathbb{E}(Y^2) = \int_0^1 y^2 \frac{7}{2} y^{5/2} dy = \frac{7}{11}$$

$$\mathbb{V}(Y) = \frac{7}{11} - \left(\frac{7}{9}\right)^2$$

$$\begin{aligned} \mathbb{E}(XY) &= \int_{x=-1}^{x=1} \int_{y=x^2}^{y=1} xy \frac{21}{4} (x^2 y) dy dx \\ &= \int_{-1}^1 \frac{21}{4} x^3 \frac{y^3}{3} \Big|_{x^2}^1 dx = 0 \end{aligned}$$

$$\text{Cov}(X, Y) = 0 - (0) \left(\frac{7}{9}\right) = 0$$

Since ρ is the covariance divided by the standard deviations, $\rho = 0$ as well.

This brings up an interesting fact, the correlation is zero, but X and Y are not independent. In order for this to be true, $f(x, y) = f_X(x)f_Y(y)$ for every point in the region.

Pick $x = 0.5$ and $y = 0.25$. Here, x is clearly between -1 and 1 and $x^2 = 0.25 \leq 1$.

$$f_X(x) = \frac{21}{8} x^2 (1 - x^4)$$

$$f_X(0.5) = \frac{21}{8} (0.5) (1 - 0.5^4) = \frac{21}{8} (0.25) (0.9375) \approx 0.6152$$

$$f_Y(y) = \frac{7}{2} y^{5/2}$$

$$f_Y(0.25) = \frac{7}{2} (0.25)^{5/2} \approx 0.1094$$

$$f(0.5, 0.25) = \frac{21}{4} (0.5)^2 (0.25) \approx 0.3281$$

$$f_X(0.5)f_Y(0.25) = (0.6152)(0.1094) \approx 0.0673 \neq 0.3281$$

Theorem: If X and Y are independent, then $\text{Cov}(X, Y) = \rho(X, Y) = 0$.

Note: The converse of this theorem is not true as was shown above.

Proof: Assume X and Y are independent. This means that $f(x, y) = f_X(x)f_Y(y)$.

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mu_X \mu_Y$$

$$= \int \int xy f(x, y) dx dy$$

$$\begin{aligned}
&= \int \int xy f_X(x) f_Y(y) dx dy \\
&= \int y f_Y(y) dy \int x f_X(x) dx \\
&= \mathbb{E}(Y) \mathbb{E}(X)
\end{aligned}$$

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mu_X \mu_Y = \mathbb{E}(Y) \mathbb{E}(X) - \mu_X \mu_Y = 0$$

And, if the $\text{Cov}(X, Y) = 0$, $\rho(X, Y) = 0$.